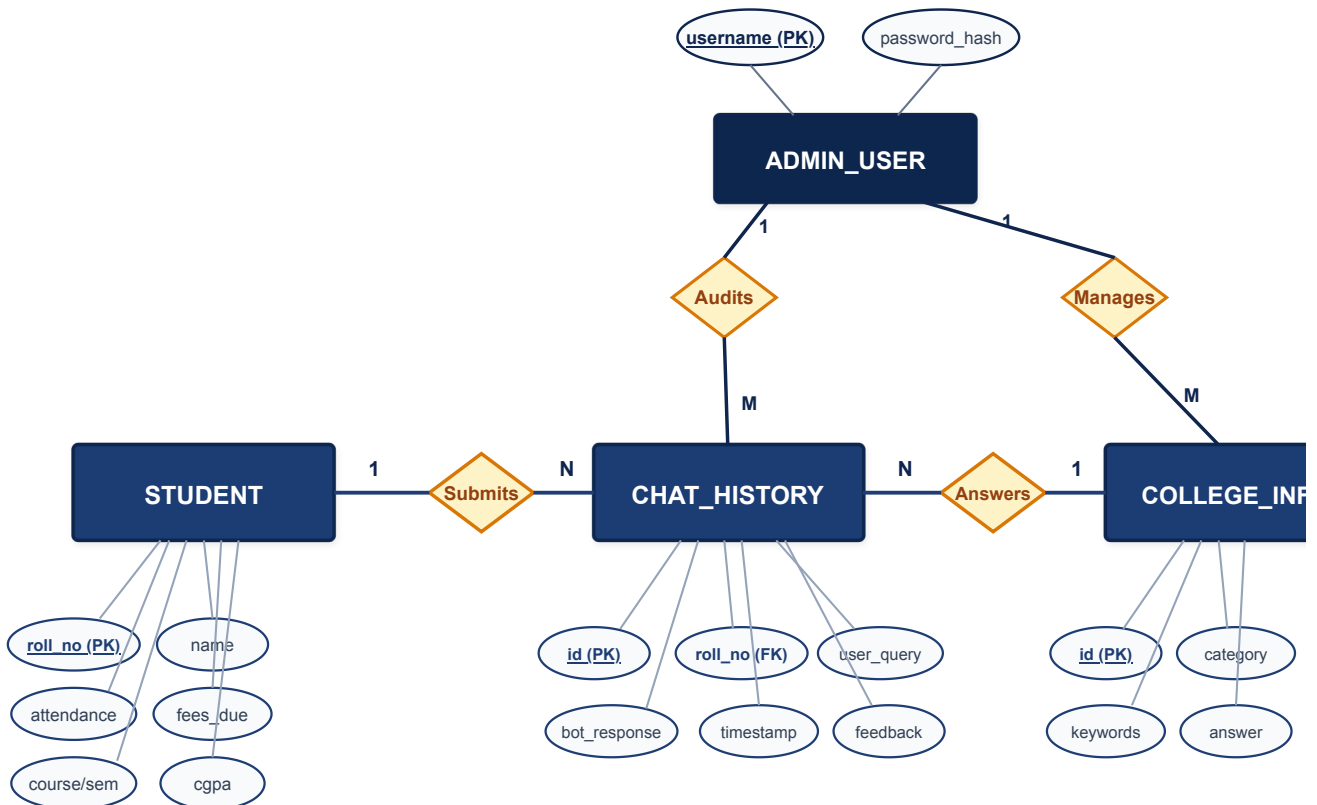


SYSTEM & UML DIAGRAMS

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7. Entity-Relationship Diagram (ERD — Database Architecture)



✓ Standard Entity-Relationship Diagram Flow Summary:

- **Left-to-Right Primary Data Flow:** STUDENT [1] submits [N] CHAT_HISTORY, which answers [N:1] from COLLEGE_INFO.
- **Top-to-Bottom Admin Control Flow:** ADMIN_USER [1] audits [M] CHAT_HISTORY and manages [M] COLLEGE_INFO FAQs.
- **Relational Integrity:** Primary Keys underlined (<u>roll_no</u>, <u>id</u>, <u>username</u>); Foreign Key marked (roll_no FK).
- Zero line crossings; clean orthogonal layout; explicit relationship diamonds with standard cardinality markers.

Fig: Entity-Relationship Diagram (ERD) — structured left-to-right relational flow with non-crossing attribute branches, primary keys, and cardinalities.